Supplement file 2

Survey to the experts of the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research (CER) - Special Interest Group: FRAMEWORK TO EVALUATE ASSESSMENT TOOLS

Dear Member,

As an expert contributor to the International Society for Pharmacoepidemiology (ISPE) Comparative Effectiveness Research (CER) Special Interest Group, we would like to invite you to take part in a short survey aimed at identifying key quality domains that:
(i) should be considered when assessing the overall quality of a nonrandomized study (NRS) in pharmacoepidemiology, and;
(ii) are not covered by existing appraisal tools for assessing the quality of NRSs.

Your answers will be used to inform the research project on an evaluation of existing tools to assess the methodological quality of pharmacoepidemiological NRSs - specifically cohort and case-control studies.

The survey will take approximately 5 minutes to complete and consists of two sections:
A. Demographic and professional information
B. Evaluation of key quality domains

In SECTION A, you will be asked to provide your demographic and professional information.

SECTION B lists key domains in assessing the overall quality of NRSs in pharmacoepidemiology.
Each key domain is presented as an item that can be part of a tool for evaluating the methodological quality of pharmacoepidemiological NRSs (cohort and case control designs). You are asked to evaluate the importance of each item, rating it from 1 (not important) to 5 (extremely important). At the end of section B, you will have the opportunity to report additional items that were not presented in the survey, are key for evaluating the methodological quality of pharmacoepidemiological NRSs and are not included in current appraisal tools.

Your answers will be kept strictly confidential and will be used only for the purposes of this research project. The results will be reported in aggregate form only and will be part of a publication in a peer-reviewed journal.

Should you have any questions on the survey or the project, please contact Elvira D'Andrea at edandrea@bwh.harvard.edu or Lydia Vinals at lydia.vinals@evidera.com.

We thank you for your collaboration.

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Section A – Demographic and professional info

Institution or Organization:

– Academia

– Industry

– Governmental Agencies

– Other (specify) _____

Professional position _____

Country _____

Section B - Evaluation of key quality domains

Please rate the following items from 1 (not important) to 5 (extremely important). Your judgment should be based on the importance that the key domains described by each item should have in tools assessing the overall methodological quality of a pharmacoepidemiologic NRS.

How important is it for a tool to assess:

1. Methods for selecting participants (selection bias)

Not important	1	2	3	4	5	Extremely important

Specifically:

1.1 Sampling strategies to correct selection bias

Not important	1	2	3	4	5	Extremely important

2.1 Inclusion and exclusion criteria of target population clearly defined

Not important	1	2	3	4	5	Extremely important

2.2 Depletion of susceptibles

Not important	1	2	3	4	5	Extremely important

2.3 Methods of selecting participants*

Not important	1	2	3	4	5	Extremely important

2.4 Other key source of selection bias not included above _____

*question added to the questionnaire of the second round of the Delphi survey based on the answers provided on the first round.

2. Definition and measurement of exposure, outcomes, covariate variables and follow-up.

Not important	1	2	3	4	5	Extremely important

Specifically:

2.1 Definition and measurement of exposure

Not important	1	2	3	4	5	Extremely important

2.2 Definition and measurement of outcomes

Not important	1	2	3	4	5	Extremely important

2.3 Definition and measurement of covariate variables

Not important	1	2	3	4	5	Extremely important

2.4 Definition and measurement of follow-up

Not important	1	2	3	4	5	Extremely important

3. Design-specific sources of bias:

Not important	1	2	3	4	5	Extremely important

Specifically:

3.1 lack of use of new user design:

Not important	1	2	3	4	5	Extremely important

3.2 lack of use of active comparator:

Not important	1	2	3	4	5	Extremely important

3.3 Time-related biases (immortal time bias or window-time bias):

Not important	1	2	3	4	5	Extremely important

3.4 Detection or surveillance bias:

Not important	1	2	3	4	5	Extremely important

3.5 Loss to follow-up bias:

Not important	1	2	3	4	5	Extremely important

3.6 Non-contemporaneous comparator bias:

Not important	1	2	3	4	5	Extremely important

3.7 Reverse causation:

Not important	1	2	3	4	5	Extremely important

3.8 Recall bias[#]:

Not important	1	2	3	4	5	Extremely important

3.10 Other critical design-specific sources of bias not mentioned above _____

[#] question removed to the questionnaire of the second round of the Delphi survey based on the answers provided on the first round.

4. Confounding:

Not important	1	2	3	4	5	Extremely important

Specifically:

4.1 Study design used to minimize the confounding:

Not important	1	2	3	4	5	Extremely important

4.2 Confounders measured and included in the analyses

Not important	1	2	3	4	5	Extremely important

4.3 Potential unmeasured confounding (addressed in the analysis through proxy analysis, IV analysis, etc.)

Not important	1	2	3	4	5	Extremely important

4.4 Other critical elements for confounding not mentioned above _____

5. Lack of appropriateness of statistical analyses (with specific mention of adjustment for causal intermediaries, and/or incorrect outcome model specification)

Not important	1	2	3	4	5	Extremely important

6. Methods for assessing statistical uncertainty in the findings (e.g., confidence intervals reported for each analysis)

Not important	1	2	3	4	5	Extremely important

7. Methods for assessing internal validity (e.g. sensitivity analysis addressing potential confounding, measurement errors or other biases)

Not important	1	2	3	4	5	Extremely important

8. Methods for assessing external validity (e.g. post-hoc subgroup analysis, comparison with other populations etc.)

Not important	1	2	3	4	5	Extremely important

9. Any other key considerations not listed above _____

GLOSSARY

SELECTION BIAS defined as a bias due to a sample selection that does not accurately reflect the target population.

DEPLETION OF SUSCEPTIBLES is defined as a phenomenon where the initiation of exposure to a drug is associated with an early increase in the incidence rate of an acute event, followed by a decrease in this incidence rate with longer duration of exposure.

NEW USER DESIGN, defined as a study design that starts following patients at the time they initiate a new drug (also known as incident user design).

ACTIVE COMPARATOR DESIGN, defined as a study design that compares the effect of the drug of interest with another drug used in clinical practice instead of non-use.

IMMORTAL TIME BIAS, defined as a bias that derives from including a period of follow-up during which, by design, outcomes cannot occur.

WINDOW-TIME BIAS, in the context of a case-control study nested in a cohort, defined as a bias that derives from the use of time-windows of different lengths between cases and controls to define time-dependent exposures.

DETECTION OR SURVEILLANCE BIAS, defined as a bias that derives from systematic differences in outcomes assessment among groups being compared.

NON-CONTEMPORANEOUS COMPARATOR BIAS, defined as a bias generated by differences in the timing of selection of comparator group(s) within a study influence exposures and outcomes resulting in biased estimates.

REVERSE CAUSATION (or reverse causality), defined as a direction of cause-and-effect contrary to a common presumption, or a two-way causal relationship between exposure and outcome.

RECALL BIAS, defined as a bias that occurs when participants do not remember previous events or experiences accurately or omit details (not for claims-based studies).

LOSS TO FOLLOW-UP BIAS, defined as a bias that occurs when there is difference in retention during the follow-up period after enrollment that are related to exposure status and outcome. It can be considered as part of selection bias.

CONFOUNDING, defined as a mixing of effects that arises when patients with different baseline risks are compared – the resulting effect measure is a mix of drug effects and risk factor effects.

ADJUSTMENT FOR CAUSAL INTERMEDIARIES, defined as adjustment for an intermediate variable (or a descending proxy for an intermediate variable) on a causal path from exposure to outcome.

INCORRECT OUTCOME MODEL SPECIFICATION, for omission of a relevant variable, inclusion of an unnecessary variable, adopting the wrong functional form, incorrect specification of the error term, uncertainty about what the true model is, reciprocal causation, etc.